

Wind Tunnel Test Procedure

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Document Information

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Revision

Version	Date	Author	Description of Changes
1.0	3 May 2026	KW	Initial Draft
1.1	20 May 2026	KW	Updated for reduced wind tunnel time

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1 Motivation

Experimental testing is required to build an accurate model for the hydrodynamic forces on a Seaglider. The 3×3 Wind Tunnel at the University of Washington provides an accessible facility to test the Seaglider in a series of dive configurations. Reynolds number similarity can be achieved in this facility.

2 Scope

This test is intended to occur on a day or days at the 3×3 Wind Tunnel at the University of Washington.

3 Test Environment/Safety/

i Test Environment

The 3×3 Wind Tunnel has, as the name suggests, a $3\text{ft} \times 3\text{ft}$ cross section in its test section. A model that is able to fit in this test section and is able to integrate with its sting balance will be small enough to be carried down by hand.

The model positioning system has no limit switches and can break itself if an incorrect move is made. A model watcher must be standing by with the E-stop button whenever the test is operational. This includes any time when the wind is on OR when the model is mounted to the balance.

ii Safety

All basic 3×3 Wind Tunnel rules and requirements will be followed. The wings and rudders may be choking hazards, and, therefore, should not be handled by children under the age of 5.

4 Roles and Responsibilities

1. _____ **Model Watcher:** Watches for structural failures in wind tunnel model or mount and controls the E-stop for the model positioning system
2. _____ **Data logger:** Logs and records data, positions model, controls dynamic pressure
3. _____ **Test Lead:** Coordinates with 3×3 staff as needed, assists data logger
4. _____ **Test Members:** Any Capstone members assisting with or observing test

5 Test Procedure

i Equipment

- ☐ Model with integrated balance mount
- ☐ Laptop with access to run log (link) and data reduction MATLAB script
- ☐ Hardware (bolts) for balance mount
- ☐ Model wings 100% condition
- ☐ Model wings 50% condition
- ☐ Model wings 25% condition
- ☐ Model rudder 100% condition
- ☐ Model rudder 50% condition
- ☐ Flash drive
- ☐ Note Taking Device
- ☐ Speed tape
- ☐ Hand tools
- ☐ Hearing protection
- ☐ Clax?
- ☐ ~~Pittsburgh Penguins merch to bribe Colin~~ Kolaches to bribe Ethan

ii Seaglider Limits

The model will be tested at the following limits. In lieu of negative AoA's, the model will be installed upside down.

Sweep	Max	Min
AoA (α)	15°	0°
Sideslip Angle (β)	10°	-10°

Table 1: Min and max values of sweeps

iii Test Scenarios and Damage Modes

1 Nominal (No Damage Modes)

1. 100% of wing attached
2. 100% of rudder attached

2 Off Nominal (Damage Modes)

1. 50% of wing attached
2. 25% of wing attached
3. 50% of rudder attached

3 AoA Sweeps

All nominal and off nominal cases will perform this AoA sweep:

A1: 0, 5, 10, 15

For all rudder damage cases, this will sweep will be repeated with the model installed upside down to simulate a negative AoA sweep.

4 Sideslip Angle Sweeps

The nominal and rudder damage cases will perform this sideslip angle sweep at all AoA's in its AoA sweep:

B1: 0, 4, 8, 10

All wing damage cases will perform this sideslip angle sweep at all AoA's in its AoA sweep:

B2: -10, -8, -4, 0, 4, 8, 10

Data is to be recorded at each point in these sideslip angle sweeps.

iv Procedure

1. _____ Install model with nominal wing box
2. _____ Run sweep A1 at each point on sweep B1, updating run log at each run
3. _____ Uninstall model, reinstall upside down
4. _____ Uninstall nominal rudder, replace with 50% remaining rudder, install model right-side up
5. _____ Run sweep A1 at each point on sweep B1, updating run log at each run
6. _____ Uninstall model, reinstall right side up, replace 0% remaining rudder with nominal rudder, uninstall nominal wing, replace with 50% remaining wing
7. _____ Run sweep A1 at each point on sweep B2, updating run log at each run
8. _____ Uninstall model, reinstall upside down
9. _____ Run sweep A1 at each point on sweep B2, updating run log at each run
10. _____ Uninstall 25% remaining wing, replace with 25% remaining wing
11. _____ Run sweep A1 at each point on sweep B2, updating run log at each run
12. _____ Uninstall model, reinstall upside down
13. _____ Run sweep A1 at each point on sweep B2, updating run log at each run
14. _____ Save data to external device
15. _____ Run data reduction program on data

6 Acceptance Criteria

1. (Yes/No) Data collected showed noticeable difference in nominal/off-nominal conditions
2. (Yes/No) Data collected showed noticeable difference between AoA and sideslip angle states
3. (Yes/No) Test was concluded safely
4. (Yes/No) Any changes to procedure or run log have been documented

7 Test Acceptance

By signing below, the UW AA Ocean Capstone Test Lead and Approver confirm that the test procedure was executed in the designated environment.

Test Lead Signature & Date

Approver Signature & Date